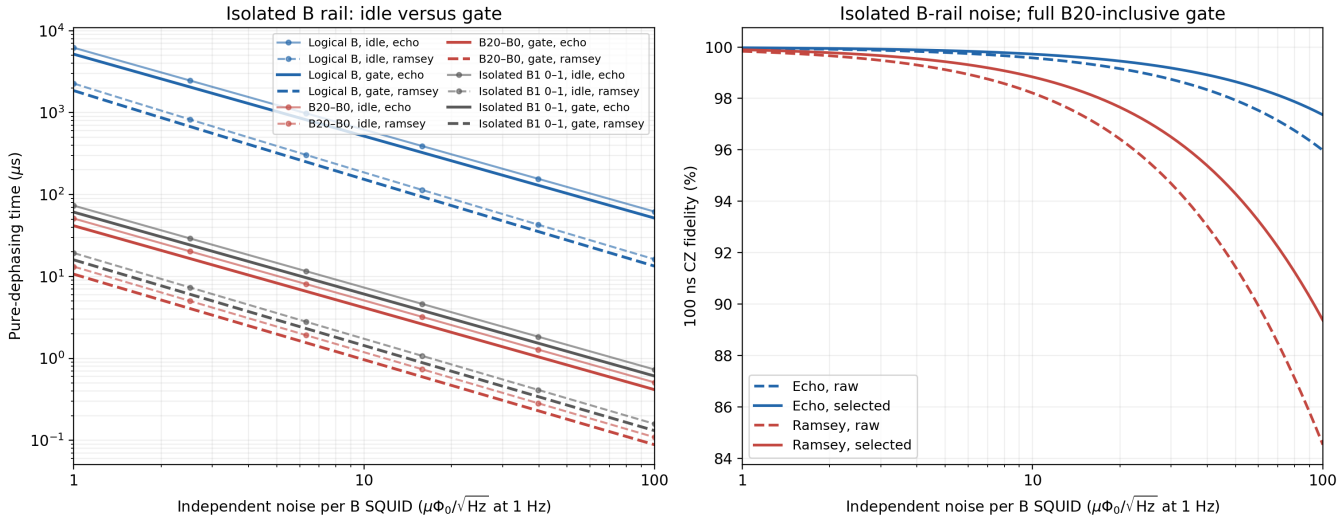
B-rail and coupler-induced dephasing

Central result: isolated B rail | independent intrinsic SQUID flux noise



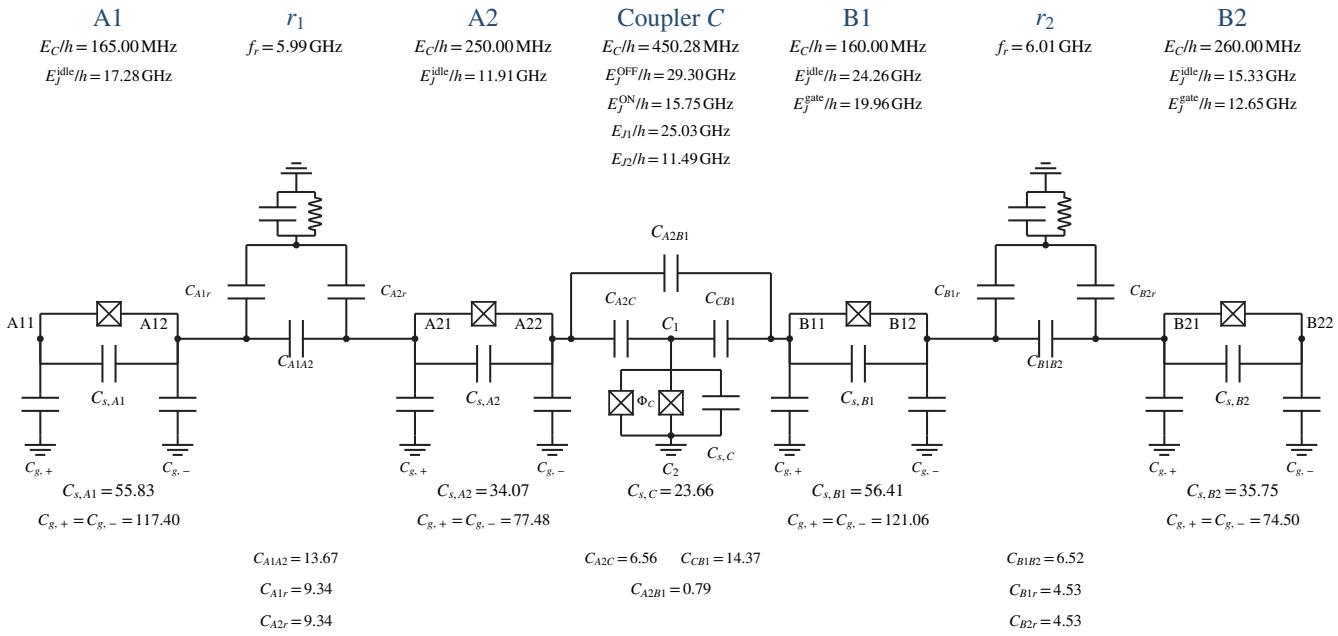
No coupler. Idle: resonant bare B1/B2 at 5.390 GHz; gate: 4.871 GHz.
Coherence: solid echo, dashed Ramsey; circles mark idle. Fidelity: exponential reference-rate model; A noiseless; no T1 or quadratic noise.

Left: logical B, B20-to-logical-B0, and isolated physical B1 0-1 pure dephasing, at idle and gate. Solid: echo; dashed: Ramsey; circles: idle. Right: 100 ns ideal CZ, including B20, with exponential rates anchored separately to echo or Ramsey. Coupler-induced noise is excluded from this central result. Independent equal-amplitude noise acts on B1 and B2; A is noiseless.

Noise per B loop	Echo raw	Echo selected	Ramsey raw	Ramsey selected
1	99.9574%	99.9725%	99.8350%	99.8934%
10	99.5765%	99.7256%	98.2114%	98.8373%
100	95.9819%	97.3619%	84.5248%	89.3685%

Noise units: $\mu\Phi_0/\sqrt{\mathrm{Hz}}$ at 1 Hz. First-order pure dephasing only; no T1, quadratic or transverse noise, gate-matrix-element fluctuations, pulse ramps or baseline coherent error. These are additional-noise budgets, not total experimental gate predictions.

1 | Circuit context and fixed inputs



All capacitances are in fF. Readout branches are contextual; decoherence calculations exclude resonator dynamics.

The schematic is the saved nominal circuit template. Readout resonators are context only; the isolated B-rail calculation contains B1 and B2 alone, and the separate coupler calculation contains B1-B2-C. No new seven-mode calculation is claimed. Instantaneous loaded EC and EJ labels describe the nominal saved circuit, not new SQUID junction values.

Input	Value
Loaded EC/h, A1/A2/B1/B2	165 / 250 / 160 / 260 MHz
Loaded coupler EC/h	450.284990 MHz
B isolated pair idle / gate bare frequency	5.390 / 4.871277772 GHz; B1 and B2 resonant separately at each point
B rail tuning range	4.4-5.8 GHz for each asymmetric SQUID
Coupler OFF / ON frequency	9.800130603 / 7.050 GHz
Coupler SQUID tuning range	6.5-11 GHz
Gate budget	100 ns ideal logical10-to-B20 cycle; cyclic g=5 MHz
Noise convention	Independent per-loop 1/f ASD at 1 Hz; infrared cutoff 1 Hz

2 | Isolated B-rail details

Retain the nominal loaded EC and intra-pair charge coefficient J = -27.2512 MHz when omitting C. Set both bare B frequencies equal at each operating point. Logical B is the splitting between lower/upper one-excitation pair eigenstates. B20 is the hybridized two-excitation auxiliary associated with physical B1=2,B2=0. B20-B0 is the auxiliary coherence across the B component of the intended gate transition; B20-ground is omitted from the central figures.

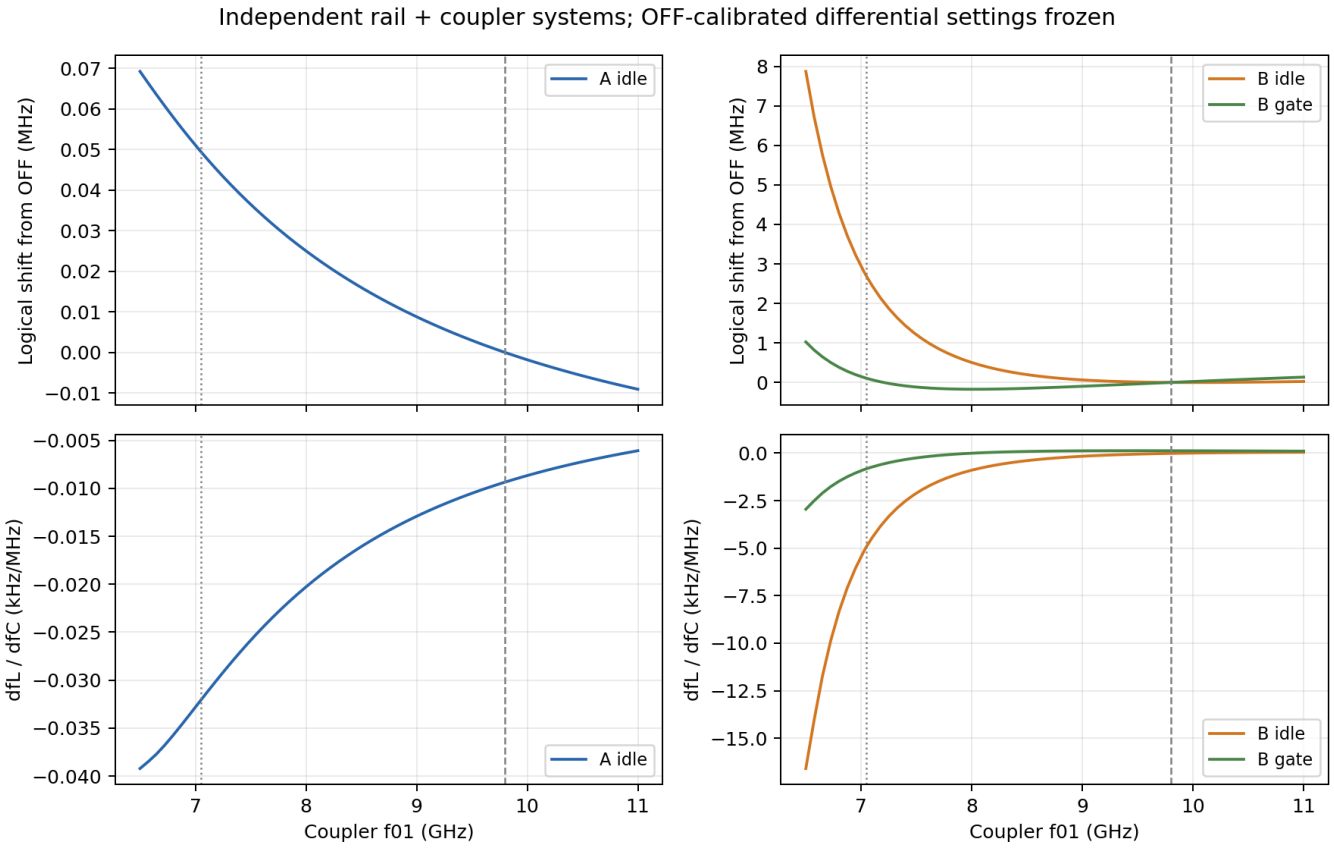
SQUID design	B1	B2
Loaded EC/h (MHz)	160.00	260.00
Large junction EJ/h (GHz)	22.04	14.12
Small junction EJ/h (GHz)	5.75	3.61
Asymmetry	0.5861	0.5932

At amplitude 10 per loop	Position	Echo Tphi (us)	Ramsey Tphi (us)
Logical B	idle	618.89	185.99
B20-B0	idle	5.09	1.19
Isolated B1 0-1	idle	7.32	1.73
Logical B	gate	516.06	153.16
B20-B0	gate	4.16	0.96
Isolated B1 0-1	gate	6.09	1.43

The isolated B1 reference removes B2 and C, retains loaded EC=160 MHz and its 4.4-5.8 GHz SQUID, and uses the corresponding bare B1 frequency. This is a fixed-loaded-EC reference, not a geometric capacitor-removal simulation. Pair logical splitting is 90.00 MHz at idle and 81.33 MHz at gate with the same physical coupling.

At resonance fL approximately equals sqrt(Delta squared + 4g squared). First-order differential-detuning sensitivity cancels, while EJ dependence of charge matrix elements modulates g. Independent B flux noise therefore still dephases the logical splitting; the auxiliary is much more sensitive. Common electrical ground alone does not imply correlated magnetic flux noise.

3 | Separate coupler-induced dephasing

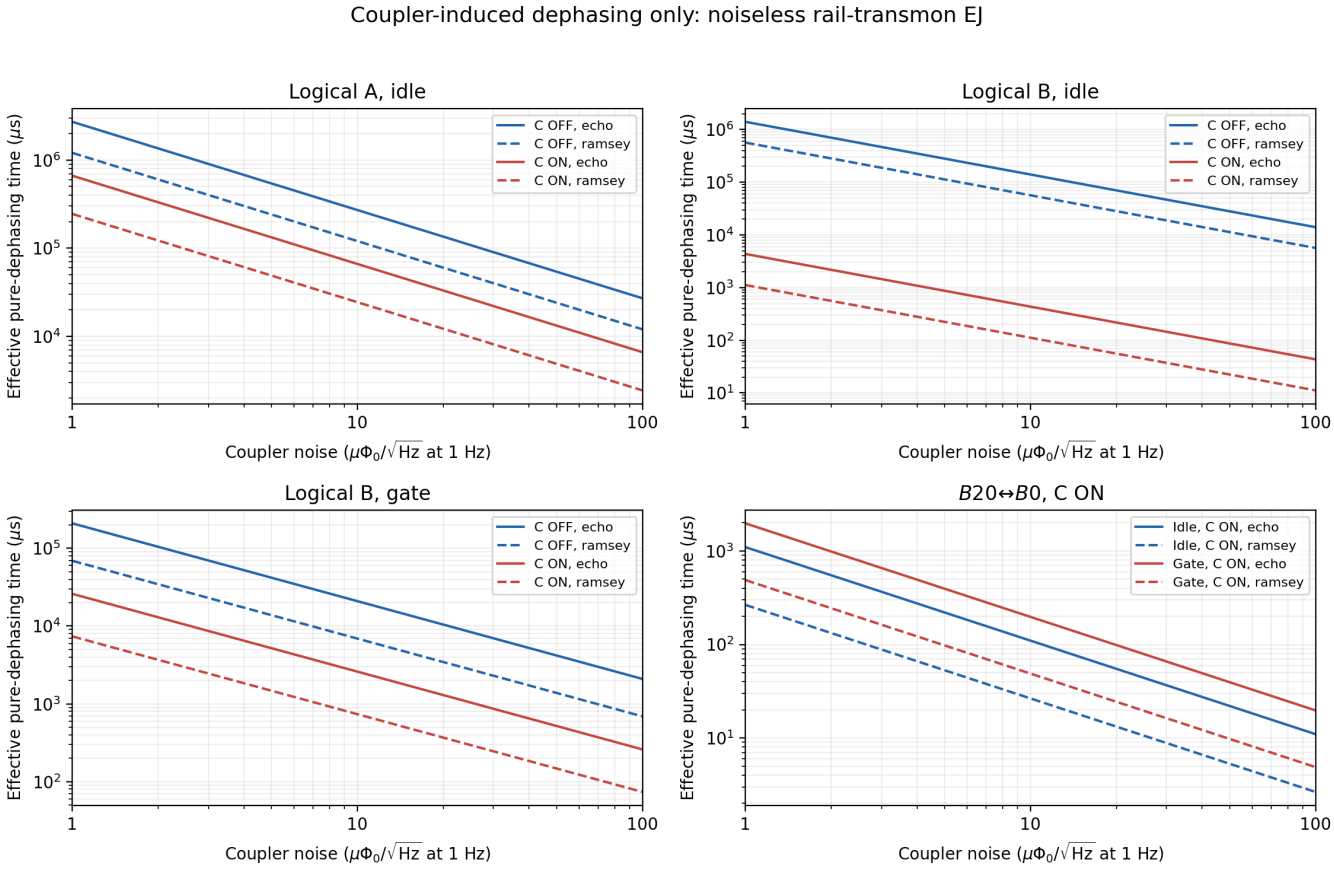


For this separate estimate, B1/B2 EJ are noiseless. Differential rail bias is calibrated at idle with C OFF, then frozen during the common move to gate frequency. Only C EJ is swept. This retained-calibration convention differs from independently resonating the isolated B pair at every position. The comparison below is therefore a comparison of the stated approximations, not an identical-control noise subtraction.

B position / C bias	Logical B echo (us)	B20-B0 echo (us)
idle / OFF	14010.51	--
idle / ON	43.47	10.99
gate / ON	259.58	19.74

All entries use coupler noise amplitude 100 $\mu\Phi_0/\sqrt{\text{Hz}}$. B20 values are shown at C ON where the auxiliary is relevant to the gate; the idle/OFF row is retained only as a logical idle reference. Coupler-induced phase noise is not a population lifetime. The SQUID large/small junction EJ/h are 25.03/11.49 GHz, with asymmetry 0.37090.

3a | Coupler-induced dephasing times



Echo/Ramsey reference times at amplitude 100, scaled inversely with amplitude for the exponential-rate model.
Ramsey cutoff logarithm held at the reference value; no rail flux noise or T1.

This figure shows the missing DEPHASING-TIME result, separately from the frequency-shift and sensitivity plot. Rail-transmon EJ values are fixed and noiseless. Only coupler flux noise contributes. Logical A is at idle; logical B is shown at idle and gate, with C OFF/ON. The gate-relevant B20-B0 auxiliary coherence is shown at C ON for both B positions. B20-OFF coherence is omitted.

All times are in us, with noise amplitude 1-100 $\mu\Phi_0/\sqrt{\mathrm{Hz}}$ on logarithmic axes. Solid lines: echo-derived; dashed: Ramsey-derived. To match the adopted exponential-reference convention, each time is scaled as Tphi(A)=Tphi(100)*100/A. Ramsey values keep the cutoff-dependent logarithm at its amplitude-100 reference; these are effective exponential-model times, not newly solved Gaussian 1/f Ramsey envelopes at every amplitude. The independent rail-noise central figure retains its separately stated Ramsey integration convention.

4 | Relative size, interpretation and checks

Gate-point coherence, A=100	B rail alone Tphi (us)	C noise Tphi (us)	C rate / rail rate
Logical B	51.61	259.58	0.1988
B20-B0	0.42	19.74	0.0211

Size of the coupler contribution: at equal assumed per-loop amplitudes, C-induced dephasing is about 47 times weaker than isolated-rail noise for the gate-relevant B20-B0 coherence, and about 5 times weaker for logical B. Thus an order-of-magnitude claim is justified for the auxiliary coherence but not uniformly for logical B. The added combined-noise decoherence and fidelity sections have been removed; their numerical records remain archived.

Exponential gate model

Every central fidelity curve includes B20. Use the five states logical00,01,10,11 and |gA,B20>. Physical B1/B2 flux derivatives of B0/B1/B20 determine their correlated phase shifts. For each source-family coherence, either echo or Ramsey 1/e time defines an effective exponential rate. The source rate matrix is checked for complete positivity, then the 25-dimensional density-matrix propagator is solved. Selected fidelity is average raw fidelity divided by average logical survival, not an unweighted mean of input-normalized fidelities.

Echo: Tphi=1/[pi sqrt(2ln2) a], with frequency-noise ASD a. Ramsey times are solved at each amplitude from the finite-cutoff Gaussian 1/f filter integral, using cutoff 1 Hz. Substituting either reference into an exponential model is phenomenological; it does not reproduce the original Gaussian short-time envelope. First-order sensitivities are used consistently. The earlier quadratic Ramsey calculation remains an archived separate approximation.

Checks and scope

Full-cosine local transmons and charge interactions are used. Isolated 7-to-8-level logical-frequency changes are below 0.03 Hz. Coupler subsystem audits preserve quoted rounded times. No readout, decoherence-induced transverse transitions, T1, full-device gate retuning or pulse ramps are included. The fabrication robustness campaign is a separate coherent-device calculation and does not include these dephasing errors.

Plot/data folders: output/B-rail-intrinsic-flux-noise-20261030 and output/independent-rail-coupler-noise-20261030. Ramsey/echo revision records remain archived under output/combined-B-coupler-flux-noise-20261030; their combined-noise plots are no longer part of this report.